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FURTHER MATHEMATICS

9231/31

Paper 3 Further Mechanics

October/November 2021

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

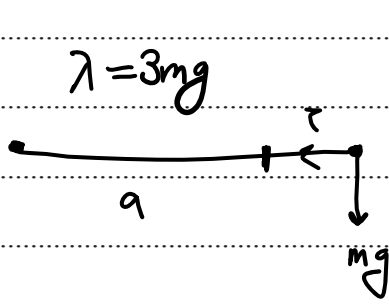
This document has **16** pages. Any blank pages are indicated.

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- 1 One end of a light elastic string, of natural length a and modulus of elasticity $3mg$, is attached to a fixed point O on a smooth horizontal plane. A particle P of mass m is attached to the other end of the string and moves in a horizontal circle with centre O . The speed of P is $\sqrt{\frac{4}{3}ga}$.

Find the extension of the string.

[4]



$$\lambda = 3mg$$

$$T = \frac{mv^2}{r}$$

$$= \frac{m \frac{4}{3}ga}{(a+x)}$$

$$= \frac{4mg a}{3(a+x)}$$

Also:

$$T = \frac{\lambda x}{a}$$

$$= \frac{3mgx}{a}$$

$$\hookrightarrow \frac{3mgx}{a} = \frac{4mg a}{3(a+x)}$$

$$9(x+a)(x) = 4a^2$$

$$9x^2 + 9ax = 4a^2$$

$$9x^2 + 9ax - 4a^2 = 0$$

$$x = \frac{-9a \pm \sqrt{81a^2 - 4(9)(-4a^2)}}{18}$$

$$= \frac{-9a \pm \sqrt{225a^2}}{18} = \left(\frac{a}{3} \right)$$

(as $x > 0$)

- 2 A particle P of mass m kg moves along a horizontal straight line with acceleration $a \text{ ms}^{-2}$ given by

$$a = \frac{v(1-2t^2)}{t},$$

where $v \text{ ms}^{-1}$ is the velocity of P at time t s.

- (a) Find an expression for v in terms of t and an arbitrary constant.

[3]

$$\frac{dv}{dt} = \frac{v(1-2t^2)}{t}$$

$$\int \frac{dv}{v} = \int \frac{1-2t^2}{t} dt$$

$$\ln|v| = \int \frac{1}{t} dt - \int 2t dt$$

$$\ln|v| = \ln|t| - t^2 + c$$

$$|v| = e^{\ln t - t^2 + c}$$

$$|v| = t(e^{c-t^2}) = Ate^{-t^2}$$

- (b) Given that $a = 5$ when $t = 1$, find an expression, in terms of m and t , for the horizontal force acting on P at time t .

[3]

$$@ t=1, v=-5$$

$$5 = A(1)(e^{-1})$$

$$A = 5e$$

$$\hookrightarrow |v| = 5te^{-t^2}$$

$$F = -5me^{1-t^2}(1-2t^2)$$

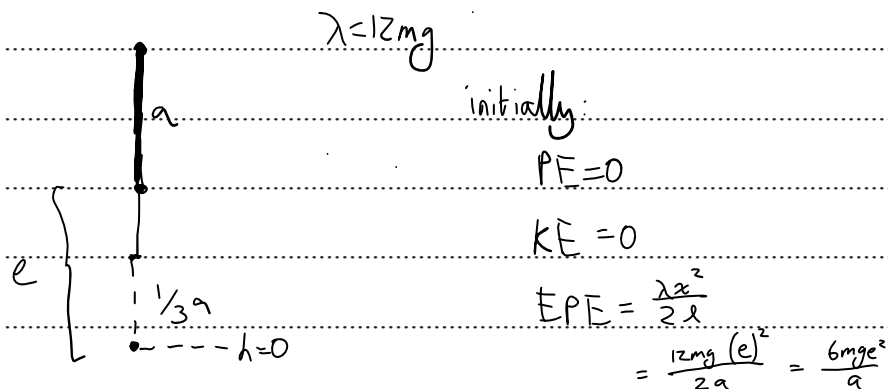
$$v = -5te^{-t^2}$$

$$a = \frac{-5te^{-t^2}(1-2t^2)}{t} \Rightarrow a = -5e^{1-t^2}(1-2t^2)$$

- 3 A light elastic string has natural length a and modulus of elasticity $12mg$. One end of the string is attached to a fixed point O . The other end of the string is attached to a particle of mass m . The particle hangs in equilibrium vertically below O . The particle is pulled vertically down and released from rest with the extension of the string equal to e , where $e > \frac{1}{3}a$. In the subsequent motion the particle has speed $\sqrt{2ga}$ when it has ascended a distance $\frac{1}{3}a$.

Find e in terms of a .

[6]



Finally:

$$PE = mg\left(\frac{a}{3}\right) = \frac{mga}{3}$$

$$KE = \frac{1}{2}m(\sqrt{2ga})^2 = mga$$

$$EPE = \frac{\lambda x^2}{2l} = \frac{12mg(e - \frac{1}{3}a)^2}{2(a)}$$

$$= \frac{6mg}{a} \left(e - \frac{1}{3}a\right)^2$$

By CoE:

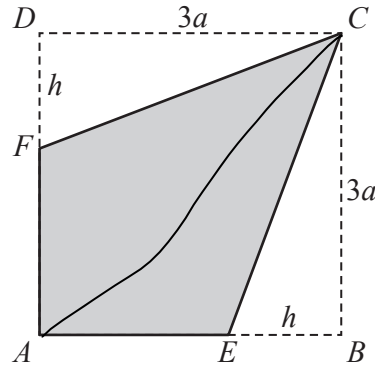
$$\frac{6mg}{a} e^2 = \frac{6mg}{a} \left(e - \frac{1}{3}a\right)^2 + \frac{4mga}{3}$$

$$\frac{6e^2}{a} = \frac{6}{a} \left(e^2 + \frac{1}{9}a^2 - \frac{2}{3}ea\right) + \frac{4a}{3}$$

$$\frac{6e^2}{a} = \frac{6e^2}{a} + \frac{2}{3}a - 4e + \frac{4a}{3}$$

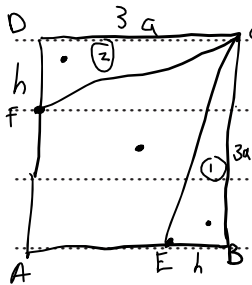
$$4e = 2a$$

$$e = \frac{a}{2}$$



A uniform lamina $AECF$ is formed by removing two identical triangles BCE and CDF from a square lamina $ABCD$. The square has side $3a$ and $EB = DF = h$ (see diagram).

- (a) Find the distance of the centre of mass of the lamina $AECF$ from AD and from AB , giving your answers in terms of a and h . [5]



→ this shape is a square - 2 triangles

$$\text{COM (square)} : (1.5a, 1.5a)$$

$$\text{COM (1)} : (3a - \frac{h}{3}, a)$$

$$\text{COM (2)} : (a, 3a - \frac{h}{3})$$

Considering moments about AB:

$$(\bar{y})(9a^2 - 3ah) = \left(\frac{3ah}{2}\right)(a) + \left(\frac{3ah}{2}\right)\left(3a - \frac{h}{3}\right) + (9a^2)(1.5a)$$

$$(\bar{y})(6)(3a - h) = \frac{-3a^2h}{2} + \frac{ah^2}{2} - \frac{9a^2h}{2} + \frac{27a^3}{2}$$

$$(\bar{y})(6)(3a - h) = -3ah + h^2 - 9ah + 27a^2$$

$$\bar{y} = \frac{h^2 - 12ah + 27a^2}{6(3a - h)} \quad \text{from both AB \& AD}$$

$$= \frac{(3a - h)(9a - h)}{6(3a - h)}$$

$$= \frac{9a - h}{6} \quad \rightarrow \bar{x} = \bar{y} \quad (\text{by symmetry})$$

The lamina $AE CF$ is placed vertically on its edge AE on a horizontal plane.

- (b) Find, in terms of a , the set of values of h for which the lamina remains in equilibrium. [3]

$$\frac{9a-h}{6} \leq 3a-h$$

$$9a-h \leq 18a-6h$$

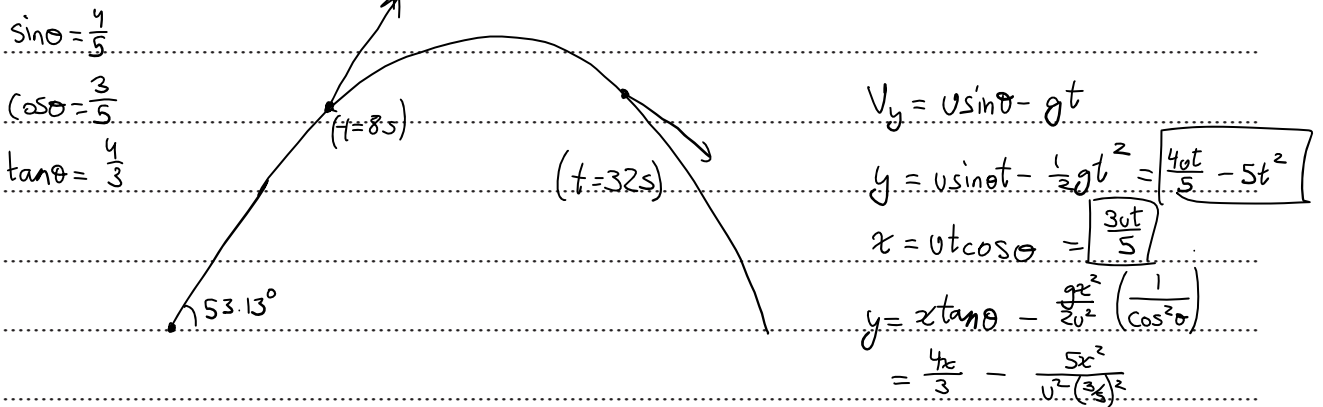
$$5h \leq 9a$$

$$h \leq \frac{9a}{5}$$

- 5 A particle P is projected from a point O on a horizontal plane and moves freely under gravity. Its initial speed is $u \text{ ms}^{-1}$ and its angle of projection is $\sin^{-1}(\frac{4}{5})$ above the horizontal. At time 8 s after projection, P is at the point A . At time 32 s after projection, P is at the point B . The direction of motion of P at B is perpendicular to its direction of motion at A .

Find the value of u .

[7]



@ $t=8$:

$$x = \frac{24u}{5}$$

$$y = \frac{32u}{5} - 320$$

$$\left[-\frac{4x}{3} - \frac{125x^2}{9u^2} \right]$$

$$\frac{dy}{dx} = \frac{4}{3} - \frac{250x}{9u^2}$$

@ $t=32$:

$$x = \frac{96u}{5}$$

$$y = \frac{128u}{5} - 5120$$

@ $t=8s$, $x = \frac{24u}{5}$

$$\frac{dy}{dx} = \frac{4}{3} - \frac{250 \left(\frac{24u}{5} \right)}{9u^2} = \frac{4}{3} - \frac{400}{3u}$$

@ $t=32s$, $x = \frac{96u}{5}$

$$\frac{dy}{dx} = \frac{4}{3} - \frac{250 \left(\frac{96u}{5} \right)}{9u^2} = \frac{4}{3} - \frac{1600}{3u}$$

$$\frac{\frac{4}{3} - \frac{400}{3u}}{-1} = \frac{4}{3} - \frac{1600}{3u}$$

$$1 = \left(\frac{400}{3v} - \frac{4}{3} \right) \left(\frac{4}{3} - \frac{1600}{3v} \right)$$

$$1 = \left(\frac{400-4v}{3v} \right) \left(\frac{4v-1600}{3v} \right)$$

$$\begin{aligned} 9v^2 &= (400-4v)(4v-1600) \\ &= 4(100-v)(v-400)(4) \end{aligned}$$

$$\frac{9}{16} v^2 = -v^2 + 500v - 40000$$

$$0 = \frac{25v^2}{16} - 500v + 40000$$

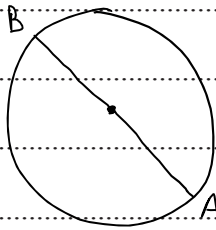
$$v = 160 \text{ ms}^{-1}$$

- 6 A particle P , of mass m , is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The particle P moves in complete vertical circles about O with the string taut. The points A and B are on the path of P with AB a diameter of the circle. OA makes an angle θ with the downward vertical through O and OB makes an angle θ with the upward vertical through O . The speed of P when it is at A is $\sqrt{5ag}$.

The ratio of the tension in the string when P is at A to the tension in the string when P is at B is $9 : 5$.

- (a) Find the value of $\cos \theta$.

[6]



Let initial speed $= U$

$$T_A = \frac{mv^2}{a} - 2mg + 3mg \cos \theta$$

$$T_B = \frac{mv^2}{a} - 2mg + 3mg (\cos \theta + 180^\circ)$$

$$= \frac{mv^2}{a} - 2mg - 3mg \cos \theta$$

By CoE: (@A):

$$\frac{mv^2}{r} = \frac{mv^2}{a} - 2mg + 2mg \cos \theta$$

$$\frac{5ga}{a} = \frac{v^2}{a} - 2g + 2g \cos \theta$$

$$\textcircled{1} \quad \frac{v^2}{a} = 7g - 2g \cos \theta$$

$$9T_B = 5T_A$$

$$\frac{9mv^2}{a} - 18mg - 27mg \cos \theta = \frac{5mv^2}{a} - 10mg + 15mg \cos \theta$$

$$\frac{4mv^2}{a} - 8mg - 42mg \cos \theta = 0$$

$$\frac{v^2}{a} - 2g - 10.5g \cos \theta = 0$$

$$\frac{v^2}{a} = 2g + 10.5g \cos \theta \quad \textcircled{2}$$

Equating ① & ②:

$$7g - 2g \cos \theta = 2g + 10.5g \cos \theta$$

$$5g = 12.5g \cos \theta$$

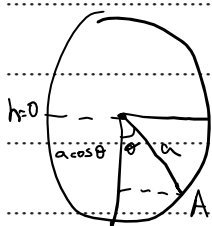
$$\cos \theta = \frac{5}{12.5}$$

$$\cos \theta = \frac{2}{5}$$

(b) Find, in terms of a and g , the greatest speed of P during its motion.

[2]

Taking point A:



$$PE = mgh$$

$$= mg(-a \cos 50^\circ)$$

$$= -mg a \cos 50^\circ = \frac{-2mga}{5}$$

$$KE = \frac{1}{2} mv^2$$

$$= \frac{1}{2} m(5ga)$$

$$= \frac{5}{2} mga$$

$$\therefore \text{Total energy} = 2.1mga$$

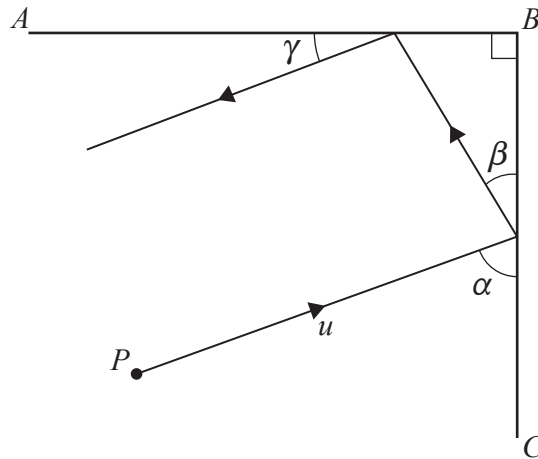
→ Max KE achieved when $h = -a$

$$\hookrightarrow PE = -mga$$

$$KE = 3.1mga \quad (\text{CoE})$$

$$\hookrightarrow = \frac{1}{2} m(\sqrt{6.2ga})^2$$

$$\hookrightarrow v_{\max} = \sqrt{6.2ga} \quad \text{ms}^{-1}$$



The smooth vertical walls AB and CB are at right angles to each other. A particle P is moving with speed u on a smooth horizontal floor and strikes the wall CB at an angle α . It rebounds at an angle β to the wall CB . The particle then strikes the wall AB and rebounds at an angle γ to that wall (see diagram). The coefficient of restitution between each wall and P is e .

- (a) Show that $\tan \beta = e \tan \alpha$.

[3]

Upon collision, only v_x is affected

$$\rightarrow v_x = u \sin \alpha$$

$$v_x = e u \sin \alpha$$

$$v_y = v_y = u \cos \alpha$$

$$\tan \beta = \frac{v_x}{v_y} = \frac{e u \sin \alpha}{u \cos \alpha} = e \tan \alpha$$

- (b) Express γ in terms of α and explain what this result means about the final direction of motion of P .

[4]

lets take

As established

$$v_{\text{initial}} = u$$

$$v_x = u \sin \alpha$$

$$v_{\text{middle}} = v$$

$$v_x = e u \sin \alpha$$

$$v_{\text{final}} = w$$

v_x is unaffected by the vertical collision

$$\therefore w_x = e u \sin \alpha$$

$$v_y = v \cos d$$

v_y is unaffected by horizontal collisions.

$$v_y = v \cos d$$

$$w_y = e v \cos d$$

$$\tan Y = \frac{w_y}{w_x} = \frac{e v \cos d}{e v \sin d}$$

$$\tan Y = \frac{1}{\tan d}$$

$$Y = 90 - d$$

As a result of the two impacts the particle loses $\frac{8}{9}$ of its initial kinetic energy.

(c) Given that $\alpha + \beta = 90^\circ$, find the value of e and the value of $\tan \alpha$.

[4]

$$\text{final speed} = \sqrt{w_x^2 + w_y^2}$$

$$= \sqrt{e^2 v^2 \sin^2 d + e^2 v^2 \cos^2 d}$$

$$= e v$$

$$\text{Final KE} = \frac{1}{2} m e^2 v^2$$

(taking mass = m)

$$\text{Initial KE} = \frac{1}{2} m v^2$$

$$\frac{1}{9} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m e^2 v^2$$

$$e^2 = \frac{1}{9}$$

$$\boxed{e = \frac{1}{3}}$$

$$e \tan d = \tan \beta$$

$$\boxed{\frac{1}{3} \tan d = \tan \beta}$$

$$\alpha + \beta = 90$$

$$\hookrightarrow \beta = 90 - d$$

$$\frac{1}{3} \tan d = \tan (90 - d)$$

$$\frac{1}{3} \tan d = \frac{1}{\tan d}$$

$$\hookrightarrow (\tan d)^2 = 3 \Rightarrow \boxed{\tan d = \sqrt{3}} \Rightarrow \boxed{\alpha = 60^\circ}$$

[illegible]

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